

Catherine Zhang

425-504-9803 | cat.zhang2006@gmail.com | [linkedin.com/in/catherinezhangg/](https://www.linkedin.com/in/catherinezhangg/) | github.com/Claohuu | claohuu.github.io

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science, GPA: 3.66/4.0

Sep. 2024 – June 2028 (Expected)

- *Coursework: Artificial Intelligence, Computer Graphics, Computer Vision, Data Structures and Algorithms, Hardware/Software Interface, Computational Probability*

EXPERIENCE

Gameplay Programming Intern

June 2026 – September 2026

Aggro Crab Games | *C#, Unity, Blender, Shader Graph*

Seattle, WA

- Sole programming intern on an unannounced title. Designed a centralized entity manager that distributes visibility checks across staggered coroutine cycles, amortizing per-entity work over successive frames, allowing for a **25%** increase in framerate.
- Developed a custom Unity editor tool for procedurally placing prefabs along a spline, sampling positions along the curve at configurable density with surface-snapping raycasts and gizmo previews, replacing manual hand-placement.
- Animated environmental assets through Blender shape keys, Shader Graph vertex-offset shaders, and rigged Unity Animator clips, and added C# interaction logic and custom particle effects.
- Playtested and reviewed builds for a major PEAK update (**10M+** copies sold), identifying and reporting gameplay bugs and giving feedback that informed design decisions ahead of release.

Undergraduate Research Assistant

August 2025 – December 2025

The Machine Agency at UW | *JavaScript, G-Code, 3D Printing*

Seattle, WA

- Applied p5.fab for direct G-code machine control across 3D print workflows, stress-testing serial communication and specifying toolpath streaming fixes compatible with **4+** industry standard printers.
- Scripted custom toolpaths bypassing traditional CAD/CAM software, tuning low-level machine parameters to produce non-planar tactile graphics for visually impaired learners.

Firmware Developer

Oct. 2024 – June 2025

Washington Superbike | *C++, Arduino, Raspberry Pi Pico, Teensy, CAN Bus*

Seattle, WA

- Designed a Hardware-in-the-Loop (HIL) system using Arduino, Raspberry Pi Pico, and Teensy with C++ to validate motor controller performance in electric motorcycles.
- Implemented CAN bus communication to simulate fault scenarios including battery overheating and precharge circuit failures, reducing undetected fault conditions by **~70%** prior to physical motor controller testing.

Software Engineering Intern

Aug. 2023 – July 2024

Fair Worlds | *JavaScript, HTML, 8th Wall.js, Adobe Aero*

Seattle, WA

- Engineered geospatial web AR experiences in 8th Wall.js for the Seattle Art Tour: 3D-scanned **50+** public installations, animated assets in Adobe Aero, and implemented image recognition for AR navigation.

PROJECTS

Singularituna | *Lead SWE* | *Unity, C#, Adobe Suite, Shader Graph*

September 2026

- Designed a physics-driven interaction system using Unity's Input System, with event hooks decoupling audio and VFX from core gameplay logic, routed through a self-registering component dispatcher.
- Designed **15+** modular behavior components across multiple entity types, and balanced competing resource systems through a shared pipeline routing collected resources between uses.

The Mirrors | *Lead SWE* | *Unity, C#, Adobe Suite*

December 2025 – March 2026

- Co-developed a horror-mystery game in Unity built around an enemy visible only through mirror reflections, requiring reflected geometry to stay consistent with the player's real-time viewpoint.
- Engineered a custom ray tracing system in C# to compute mirror reflections, correcting perspective distortion caused by off-axis viewing angles and non-coplanar mirror surfaces.

TECHNICAL SKILLS

Languages: Java, C#, C/C++, JavaScript, Python, HTML/CSS, G-code, Assembly

Developer Tools: Unity, Git, VS Code, Visual Studio, Blender, Shader Graph, HLSL, IntelliJ, Arduino, Adobe Suite